Filters

- Filtering
  - Form a new image whose pixels are a combination of the original pixels
- Why?
  - To get useful information from images
    - E.g., extract edges or contours (to understand shape)
  - To enhance the image
    - E.g., to remove noise
    - E.g., to sharpen or to “enhance image”

Image filtering

- Modify the pixels in an image based on some function of a local neighborhood of each pixel

10	5	3
4	5	1
1	1	7

Local image data

Some function

	7	

Modified image data

Source: L. Zhang

Linear filtering

- One simple version of filtering: linear filtering (cross-correlation, convolution)
  - Replace each pixel by a linear combination (a weighted sum) of its neighbors
- The prescription for the linear combination is called the “kernel” (or “mask”, “filter”)

10	5	3
4	6	1
1	1	8

Local image data

0	0	0
0	0.5	0
0	1	0.5

kernel

	8	

Modified image data

Source: L. Zhang

Linear filtering-Cross-correlation

Let F be the image, H be the kernel (of size 2*k*+1 x 2*k*+1), and G be the output image

$$G[i,j] = \sum_{u=-k}^{k} \sum_{v=-k}^{k} H[u,v]F[i+u,j+v]$$

This is called a **cross-correlation** operation:

$$G = H \otimes F$$

- Can think of as a “dot product” between local neighborhood and kernel for each pixel

Linear filtering-Convolution

- Same as cross-correlation, except that the kernel is “flipped” (horizontally and vertically)

$$G[i,j] = \sum_{u=-k}^{k} \sum_{v=-k}^{k} H[u,v]F[i-u,j-v]$$

This is called a **convolution** operation:

$$G = H * F$$

- Convolution is **commutative** and **associative**

*- F*G = G*F (commutative)*

- F(G*I) = (F*G)*I (associative)*

Linear filters: examples

Original

$$*\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} =$$

Identical image

Source: D. Lowe

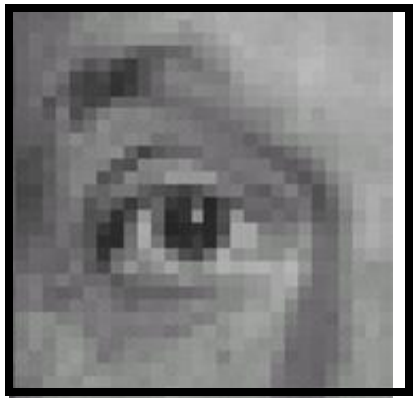
Linear filters: examples

Original

$*$

0	0	0
1	0	0
0	0	0

$=$



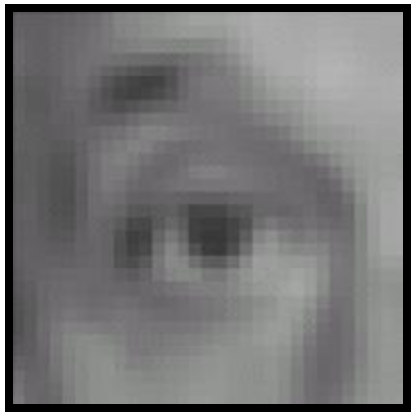
Shifted left
By 1 pixel

Source: D. Lowe

Linear filters: examples

Original

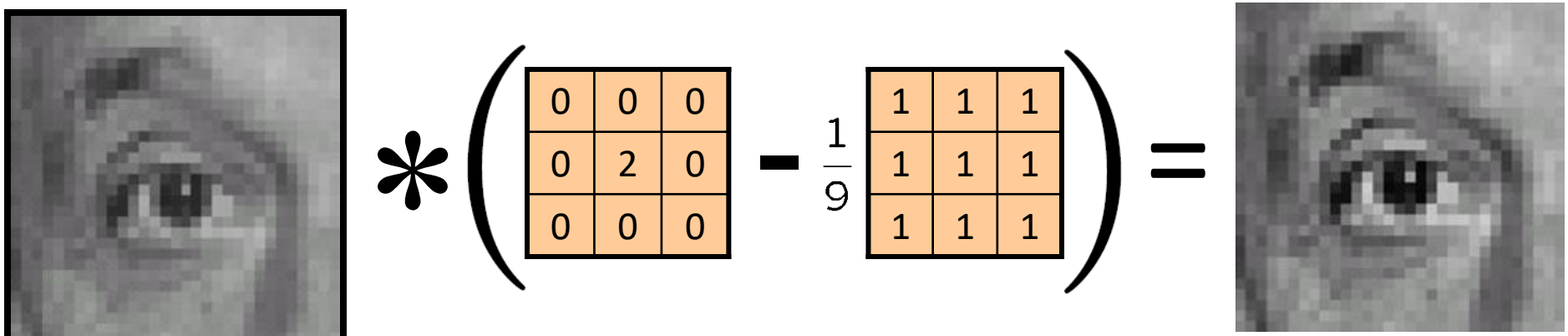
$$* \quad \frac{1}{9}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \quad =$$



Blur (with a mean filter)

Source: D. Lowe

Linear filters: examples

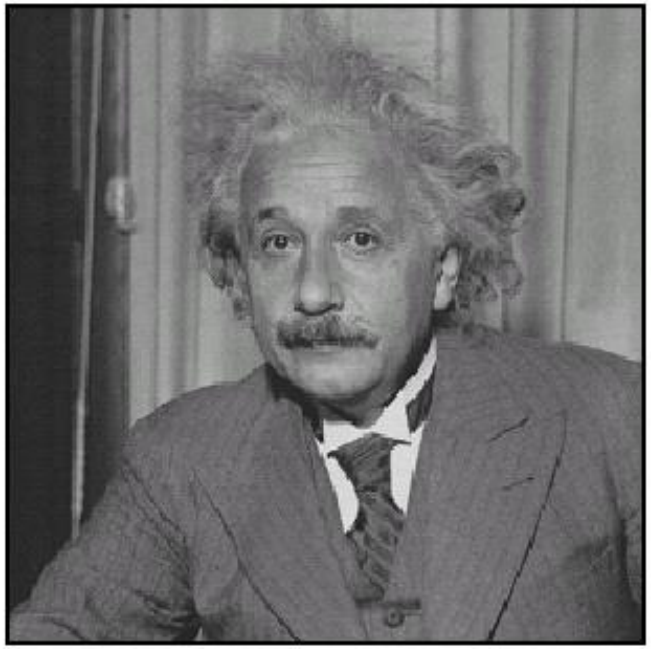


Original

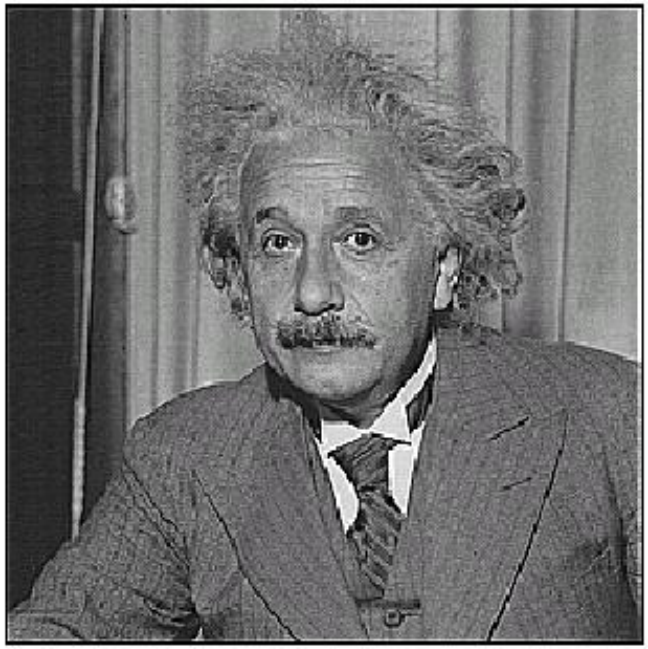
Sharpening filter

Source: D. Lowe

Sharpening



before

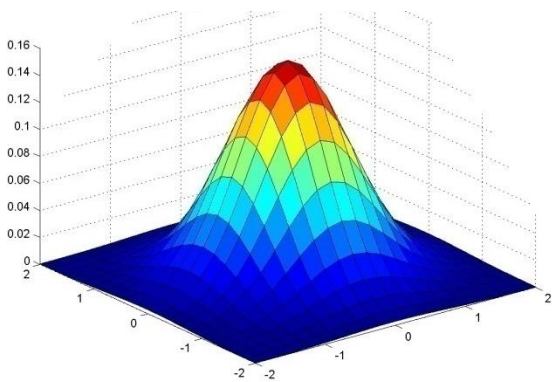


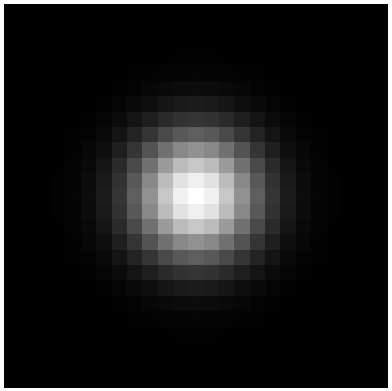
after

Source: D. Lowe

Gaussian Kernel

$$G_\sigma = \frac{1}{2\pi\sigma^2} e^{-\frac{(x^2+y^2)}{2\sigma^2}}$$





0.003	0.013	0.022	0.013	0.003
0.013	0.059	0.097	0.059	0.013
0.022	0.097	0.159	0.097	0.022
0.013	0.059	0.097	0.059	0.013
0.003	0.013	0.022	0.013	0.003

5 x 5, σ = 1

- Constant factor at front makes volume sum to 1 (can be ignored, as we should re-normalize weights to sum to 1 in any case)

Source: C. Rasmussen

Q1

- Given the following image *F* and filter *H*, answer the following questions:

$$F = \begin{bmatrix} 1 & 2 & 4 & 5 \\ 5 & 2 & 5 & 2 \\ 1 & 1 & 3 & 6 \\ 2 & 4 & 6 & 7 \end{bmatrix} \qquad H = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

(a) Show the result of $H * F$,

Q1

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Solution:

1x0+2x1+4x0+5x1+2x(-4)+5x1+1x0+1x1+3x0=5

$$H * F = \begin{bmatrix} 5 & . \\ . & . \end{bmatrix}$$

$$F = \begin{bmatrix} 1 & 2 & 4 & 5 \\ 5 & 2 & 5 & 2 \\ 1 & 1 & 3 & 6 \\ 2 & 4 & 6 & 7 \end{bmatrix} \qquad H = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

2x0+4x1+5x0+2x1+5x(-4)+2x1+1x0+3x1+6x0=-9

$$H * F = \begin{bmatrix} 5 & -9 \\ . & . \end{bmatrix}$$

$$F = \begin{bmatrix} 1 & 2 & 4 & 5 \\ 5 & 2 & 5 & 2 \\ 1 & 1 & 3 & 6 \\ 2 & 4 & 6 & 7 \end{bmatrix} \qquad H = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

Q1

- (a) Show the result of $H*F$,

Solution:

$$F = \begin{bmatrix} 1 & 2 & 4 & 5 \\ 5 & 2 & 5 & 2 \\ 1 & 1 & 3 & 6 \\ 2 & 4 & 6 & 7 \end{bmatrix} \qquad H = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$H * F = \begin{bmatrix} 5 & -9 \\ 6 & . \end{bmatrix}$$

$$F = \begin{bmatrix} 1 & 2 & 4 & 5 \\ 5 & 2 & 5 & 2 \\ 1 & 1 & 3 & 6 \\ 2 & 4 & 6 & 7 \end{bmatrix} \qquad H = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$H * F = \begin{bmatrix} 5 & -9 \\ 6 & 6 \end{bmatrix}$$

Q1

- (b) A 3X3 median filter is a filter where the output image is obtained by selecting the median of the pixel values under the filter window. Show the result of convolving the median filter with F.

$$F = \begin{bmatrix} 1 & 2 & 4 & 5 \\ 5 & 2 & 5 & 2 \\ 1 & 1 & 3 & 6 \\ 2 & 4 & 6 & 7 \end{bmatrix}$$

Q1

Solution: $F = \begin{bmatrix} 1 & 2 & 4 & 5 \\ 5 & 2 & 5 & 2 \\ 1 & 1 & 3 & 6 \\ 2 & 4 & 6 & 7 \end{bmatrix}$ 1,1,1,2,2,3,4,5,5 $\begin{bmatrix} 2 & \cdot \\ \cdot & \cdot \end{bmatrix}$

$F = \begin{bmatrix} 1 & 2 & 4 & 5 \\ 5 & 2 & 5 & 2 \\ 1 & 1 & 3 & 6 \\ 2 & 4 & 6 & 7 \end{bmatrix}$ 1,2,2,2,3,4,5,5,6 $\begin{bmatrix} 2 & 3 \\ \cdot & \cdot \end{bmatrix}$

$F = \begin{bmatrix} 1 & 2 & 4 & 5 \\ 5 & 2 & 5 & 2 \\ 1 & 1 & 3 & 6 \\ 2 & 4 & 6 & 7 \end{bmatrix}$ 1,1,2,2,3,4,5,5,6 $\begin{bmatrix} 2 & 3 \\ 3 & \cdot \end{bmatrix}$

$F = \begin{bmatrix} 1 & 2 & 4 & 5 \\ 5 & 2 & 5 & 2 \\ 1 & 1 & 3 & 6 \\ 2 & 4 & 6 & 7 \end{bmatrix}$ 1,2,2,3,4,5,6,6,7 $\begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix}$

Q1

- (c) Comment the property of median filter in comparison to mean filter learnt in class.

$$\text{Mean filter: } \frac{1}{9} \times \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Q1

- (c) Comment the property of median filter in comparison to mean filter learnt in class.

$$\text{Mean filter: } \frac{1}{9} \times \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Solution:

Median filter is a non-linear filter that could filter out the noise (pepper noise) by retaining the median value in a local area. Mean filter is a linear filter which may smooth and blur the image due to the averaging operation.